

Annotated Bibliography

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References

- [1] Francesco Asnicar, Andrew Maltez Thomas, Andrea Passerini, Levi Waldron, and Nicola Segata. Machine learning for microbiologists. *Nature Reviews Microbiology*, 22(4):191–205, Apr 2024.

This article reviews the application of machine learning techniques in microbiology, highlighting key advancements and future directions. It emphasizes the potential of machine learning to enhance our understanding of microbial communities, their functions, and their interactions with the environment. The insights gained from this review will inform my research on the use of machine learning for analyzing complex microbiome data, particularly in the context of environmental samples.

- [2] Jules M. Blais, Michael R. Rosen, and John P. Smol, editors. *Environmental Contaminants: Using Natural Archives to Track Sources and Long-Term Trends of Pollution*. Springer, Dordrecht, 2015.

This book discusses the use of natural archives, such as sediments and ice cores, to track the sources and long-term trends of environmental pollution, providing valuable insights into the history of contaminants in the environment.

- [3] Jordi. Catalan, editor. *Tracking Long-Range Atmospheric Transport of Trace Metals, Polycyclic Aromatic Hydrocarbons, and Organohalogen Compounds Using Lake Sediments of Mountain Regions*. Springer, Dordrecht, 2015.

This book chapter discusses the use of lake sediments from mountain regions as archives for tracking long-range atmospheric transport of various pollutants, including trace metals, polycyclic aromatic hydrocarbons, and organohalogen compounds. It emphasizes the importance of understanding atmospheric transport processes and their impact on remote ecosystems, particularly in the context of global environmental change. The chapter provides valuable insights for my research on the fate and transport of organic contaminants in high-altitude environments, highlighting the need for integrated

approaches that combine field measurements, laboratory analyses, and modeling efforts.

- [4] Alex Chao, Jeffrey M. Minucci, Troy M. Ferland, E. Tyler Carr, Greg Janesch, Safia Rizwan, Heather D. Whitehead, Tommy Cathey, Shirley Pu, Laura D. Brunelle, Angela L. Batt, Jon R. Sobus, and Antony J. Williams. Prioritizing chemical candidates from non-targeted analysis using metadata, spectral similarity, and hazard scoring within INTERPRET NTA. *Analytical Chemistry*, 97(29):15904–15912, 2025.

This work introduces the INTERPRET NTA computational framework, which integrates spectral similarity, metadata, and hazard scoring to prioritize chemical candidates in HRMS datasets.

- [5] Gillian L. Daly and Frank Wania. Organic contaminants in mountains. *Environmental Science & Technology*, 39(2):385–398, 2005.

This article reviews the occurrence and fate of organic contaminants in mountainous regions, highlighting the unique challenges posed by these environments for pollution monitoring and management.

- [6] Madeleine Ernst, Kyo Bin Kang, Andrés Mauricio Caraballo-Rodríguez, Louis-Felix Nothias, Joe Wandy, Christopher Chen, Mingxun Wang, et al. MolNetEnhancer: Enhanced molecular networks by integrating metabolome mining and annotation tools. *Metabolites*, 9(7):144, 2019.

MolNetEnhancer integrates molecular networking with multiple annotation tools to improve structural interpretation of complex metabolomics datasets.

- [7] Wenkai Fan. Suspect screening and semi-quantification of organic contaminants in snow and water samples on the qinghai–tibet plateau and high mountain regions. Undergraduate honors thesis, The College of Wooster, Wooster, OH, 2025. Unpublished thesis.

This thesis investigates environmental contaminants in high-altitude water and snow samples using high-resolution mass spectrometry. It highlights the importance of computational methods for molecule identification and demonstrates how suspect screening and advanced data analysis techniques can reveal pollutants in remote ecosystems.

- [8] Peter K.T. Newton S.R. et al. Fisher, C.M. Approaches for assessing performance of high-resolution mass spectrometry–based non-targeted analysis methods. *Anal Bioanal Chem*, 414:6455–6471, 2022.

This research paper provides a comprehensive overview of the current state and challenges associated with performance assessment of high-resolution mass spectrometry (HRMS)–based non-targeted analysis (NTA) methods, emphasizing the need for standardized metrics and frameworks. For my study on computational identification of unknown molecules via HRMS, particularly in analyzing organic contaminants in remote environments such as Mount

Everest, Mt. Kilimanjaro, and Tibetan rivers, this paper offers valuable insights into the evaluation criteria and reliability of NTA workflows beyond simple structural assignment. It highlights the importance of robust performance metrics, which are critical when deploying computational approaches like molecular networking and cluster analysis to interpret complex environmental data. The focus on qualitative performance metrics such as confidence in chemical identification can guide my development of algorithms for spectral clustering and molecular network construction. These strategies will ensure meaningful differentiation of unknowns and proper assessment of confidence levels. Overall, the paper situates my computational methods within the broader context of NTA validity and underscores the necessity of standardized evaluation strategies. These strategies are especially relevant when working with large, diverse datasets collected from remote or extreme locations.

- [9] Rebekah Gray, Eve Painter, Jameson W. Sprankle, Alex D. Crawford, Jillian J. Morrison, Marian Frazier, and Jennifer A. Faust. Suspect screening for pesticides in rain and snow using liquid chromatography high-resolution mass spectrometry. *Atmospheric Environment*, 291:119389, 2022.

This study employs high-resolution mass spectrometry (HRMS) for suspect screening of pesticides in environmental samples, demonstrating the method’s effectiveness in identifying contaminants in complex matrices like rain and snow.

- [10] B. M. J. Herbert, S. Villa, and C. J. Halsall. Chemical interactions with snow: Understanding the behavior and fate of semi-volatile organic compounds in snow. *Ecotoxicology and Environmental Safety*, 63(1):3–16, 2006.

This article discusses the behavior and fate of semi-volatile organic compounds in snow, emphasizing the importance of understanding these interactions for assessing environmental contamination.

- [11] Henriette Hupatz, Ingrid Rahu, Wen-Chao Wang, et al. Critical review on in silico methods for structural annotation of chemicals detected with lc/hrms non-targeted screening. *Analytical and Bioanalytical Chemistry*, 417(2):473–493, Jan 2025.

This article critically evaluates in silico methods for the structural annotation of chemicals identified through non-targeted screening techniques. It discusses the challenges and advancements in the field, providing insights into the potential of these methods for improving the identification and characterization of environmental contaminants. The review highlights the importance of integrating computational approaches with experimental data to enhance the reliability of chemical assessments in environmental studies.

- [12] Tobias Kind, Hiroshi Tsugawa, Tomas Cajka, Yan Ma, Zijuan Lai, Sajjan S. Mehta, Gert Wohlgemuth, Dinesh Kumar Barupal, Megan R. Showalter, Masanori Arita, and

Oliver Fiehn. Identification of small molecules using accurate mass MS/MS search. *Mass Spectrometry Reviews*, 37(4):513–532, 2018.

This review explains how small molecules are identified using high-resolution MS/MS techniques and spectral libraries. It highlights computational approaches and shared databases that assist in analyzing complex environmental and biological samples.

- [13] Aliz Kiss and Elke Fries. Occurrence of benzotriazoles in the rivers main, hengstbach, and hegbach (germany). *Environmental Science and Pollution Research*, 16(6):702–710, 2009.

This study investigates the occurrence of benzotriazoles in various rivers in Germany, highlighting the need for monitoring these contaminants in aquatic environments.

- [14] Zhiyuan Ma, Shunlong Meng, Xin Gao, Hui Wan, Longxiang Fang, Lei Zhu, Zhonghua Li, Yuxiao Gao, Liping Qiu, Bo Rong, and Chao Song. Human activity driven spatial heterogeneity of pollutants in a highland river basin revealed by non-target screening and improved pbt evaluation: A case study from the yalong river on the qinghai-tibet plateau,china. *Journal of Hazardous Materials*, 496:139493, 2025.

This article provides valuable insights into the complex interactions between human activities and pollutant distribution in river basins. The use of advanced screening techniques and ecological assessments highlights the importance of tailored approaches for effective environmental management. The findings underscore the need for ongoing research in this area to address emerging contaminants and their impacts on aquatic ecosystems.

- [15] K. E. Manz, A. Feerick, J. M. Braun, Y. L. Feng, A. Hall, J. Koelmel, C. Manzano, S. R. Newton, K. D. Pennell, B. J. Place, K. J. Godri Pollitt, C. Prasse, and J. A. Young. Non-targeted analysis (NTA) and suspect screening analysis (SSA): A review of examining the chemical exposome. *Journal of Exposure Science & Environmental Epidemiology*, 33(4):524–536, 2023.

This review examines the use of HRMS-based non-targeted analysis and suspect screening for identifying environmental contaminants in air, water, soil, and human samples. It highlights computational workflows and current limitations in exposome research.

- [16] Stephanie Nino-Suastegui, Eve Painter, Jameson W. Sprankle, Jillian J. Morrison, Jennifer A. Faust, and Rebekah Gray. Non-targeted analysis and suspect screening of organic contaminants in temperate snowfall using liquid chromatography high-resolution mass spectrometry. *Environmental Research*, 266:120494, 2025.

This study uses LC-HRMS and in silico analysis to identify organic contaminants deposited in snowfall across the United States, demonstrating the power of computational tools for suspect screening and compound identification.

- [17] Florent Olivon, Gwendal Grelier, Fanny Roussi, Marc Litaudon, and David Touboul. MZmine 2 data preprocessing to enhance molecular networking reliability. *Analytical Chemistry*, 89(15):7836–7840, 2017.

This study improves molecular networking workflows through data preprocessing using MZmine 2, demonstrating how careful data processing can improve clustering and compound annotation accuracy.

- [18] Robin Schmid, Daniel Petras, Louis-Felix Nothias, Mingxun Wang, Allegra T. Aron, Annika Jagels, Hiroshi Tsugawa, et al. Ion identity molecular networking for mass spectrometry-based metabolomics in the GNPS environment. *Nature Communications*, 12:3832, 2021.

This article presents Ion Identity Molecular Networking, a computational framework that links related ion species within mass spectrometry datasets to improve metabolite annotation and clustering.

- [19] Bastian Schulze, Youngjoon Jeon, Sarit Kaserzon, Amy L. Heffernan, Pradeep Dewapriya, Jake O’Brien, Maria Jose Gomez Ramos, Sara Ghorbani Gorji, Jochen F. Mueller, Kevin V. Thomas, and Saer Samanipour. An assessment of quality assurance/quality control efforts in high resolution mass spectrometry non-target workflows for analysis of environmental samples. *TrAC Trends in Analytical Chemistry*, 133:116063, 2020.

This article provides a comprehensive overview of quality assurance and quality control measures in non-target analysis workflows, highlighting the importance of robust QA/QC practices to ensure reliable results in environmental sample analysis.

- [20] Emma L. Schymanski, Junho Jeon, Rebekka Gulde, Kathrin Fenner, Matthias Ruff, Heinz P. Singer, and Juliane Hollender. Identifying small molecules via high resolution mass spectrometry: Communicating confidence. *Environmental Science & Technology*, 48(4):2097–2098, 2014.

This paper proposes a five-level confidence framework for identifying molecules in high-resolution mass spectrometry. The system is widely used in non-targeted analysis and provides standardized reporting for tentative and confirmed molecular identifications.

- [21] Mingxun Wang, Jeremy J. Carver, Vanessa V. Phelan, Laura M. Sanchez, Neha Garg, et al. Sharing and community curation of mass spectrometry data with global natural products social molecular networking. *Nature Biotechnology*, 34(8):828–837, 2016.

This article discusses the use of social molecular networking for the sharing and curation of mass spectrometry data, highlighting its potential to enhance collaboration and data accessibility in the field of natural products research.

- [22] Yue Xu, Gan Zhang, Jun Li, Paromita Chakraborty, Hua Li, and Xiang Liu. Long-range atmospheric transport of persistent organochlorinated compounds to a remote mountain site in southwestern china. *Journal of Environmental Monitoring*, 13:3119–3127, 2011.

This study investigates the long-range atmospheric transport of persistent organochlorinated compounds to a remote mountain site in southwestern China, highlighting the global implications of local pollution sources.

- [23] Weiting Ye, Jingcheng Li, and Xianfa Cai. MfGNN: Multi-scale feature-attentive graph neural networks for molecular property prediction. *Journal of Computational Chemistry*, 46(3):e70011, 2025.

This study introduces a graph neural network model for molecular property prediction using atom-level and fragment-level information. The methods demonstrate how machine learning can capture structural relationships between molecular fragments, offering insights relevant to computational identification of unknown molecules.

- [24] Weimin Zhu, Yi Zhang, Duancheng Zhao, Jianrong Xu, and Ling Wang. HiGNN: A hierarchical informative graph neural network for molecular property prediction. *Journal of Chemical Information and Modeling*, 63(1):43–55, 2023.

This paper introduces HiGNN, a graph neural network architecture designed to capture hierarchical molecular features for property prediction, demonstrating the potential of machine learning in molecular analysis.